

B03 Quick Revision

Grain, key integrity, conformed dimensions and reconciliation

Closed-book revision map

Lesson	Contract	Engineering rule
B03-L01	Process + grain	Write the one-row meaning before table roles; separate facts at different grains.
B03-L02	Keys + cardinality	One-side uniqueness and FK meaning are explicit; bridges represent real memberships, not data defects.
B03-L03	Multiple facts + dates	Use conformed dimensions and explicit date roles; do not flatten different-grain facts.
B03-L04	Reconcile + handoff	Prove raw-to-model counts/totals and hand off process, grain, keys, facts, dims, bridges, date roles and controls.

Failure-first recall

- Joining a transaction fact to a lower-grain event/line fact and then summing transaction measures can multiply those measures.
- A legitimate many-to-many membership belongs in a bridge/factless-fact at its own grain; DISTINCT is not a semantic repair.
- A monthly or weekly target is a higher-grain fact. It cannot be treated as if the target exists once per transaction.
- Role-playing dates must stay explicit: booking/order, pickup and delivery dates answer different questions.
- Reconciliation is a design control, not a dashboard cosmetic. If raw counts/totals cannot be explained, the handoff is not ready.

Move on when

With the book closed, you can state every fact grain, identify conformed dimensions/bridges/date roles, predict fan-out before joining and define the reconciliation evidence required before implementation.